COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/12/2026	**DATE SUBMITTED:**	07/15/2026
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ACTIVITY NO. 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A & B

Sections	Answer

Header File Declaration Section	`#include <iostream>`
	`// FUNCTION DECLARATION`
	`// ILO A`

Global Declaration Section	`int sumOfTwoNum(int num1, int num2);`
	`bool isGreaterThan(int a, int b);`
	`bool logicalOperation(bool x, bool y);`

Class Declaration and Method Definition Section	`class Triangle {`
	`  private:`
	`  double totalAngle, angleA, angleB, angleC;`
	`  public:`
	`  Triangle(double A, double B, double C);`
	`  void setAngles(double A, double B, double C);`
	`  const bool validateTriangle();`
	`};`

Main Function

```
int main(){
    // ILO A #2
    int x = 333;
    int y = 255;

    if (isGreaterThan(x, y)) {
        std::cout << "x is GREATER than y" << std::endl;
    } else {
        std::cout << "x is NOT GREATER than y" << std::endl;
    }

    // ILO A #3
    bool success = logicalOperation(true, false);
    std::cout << "Function status: " << std::boolalpha << success << std::endl;
    // ILO B
    Triangle set1(40,30,110);
    if(set1.validateTriangle()){
        std::cout << "The shape is a VALID Triangle \n.";
    } else {
        std::cout << "The shape is NOT A VALID Triangle \n.";
    }

    return 0;
}
```

Method Definition

```
// ILO A #1
int sumOfTwoNum(int num1, int num2){
    int sum = num1 + num2;
    return sum;
}
// ILO A #2
bool isGreaterThan(int a, int b){
    return a > b;
}
```

```
// ILO A #3
bool logicalOperation(bool x, bool y) {
    // CHANGES 1 OR 0 TO TRUE OR FALSE
    std::cout << std::boolalpha;
    std::cout << "Logical Operation " << std::endl;
    std::cout << "Values: X = " << x << ", Y = " << y << std::endl;
    // 1. NOT
    std::cout << "NOT X: " << (!x) << std::endl;
    std::cout << "NOT Y: " << (!y) << std::endl;
    // 2. AND
    std::cout << "X AND Y: " << (x && y) << std::endl;
    // 3. OR
    std::cout << "X OR Y: " << (x || y) << std::endl;
    // 4. XOR
    std::cout << "X XOR Y: " << (x ^ y) << std::endl;

    return true;
}
```

```
Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle(){
    return (totalAngle <= 180);
}
```

Table 1-1. C++ Structure Code for Answer

```
Triangle setl(40,30,110);
if(setl.validateTriangle()){
    std::cout << "The shape is a VALID Triangle \n.";
} else {
    std::cout << "The shape is NOT A VALID Triangle \n.";
}
```

```
const bool Triangle::validateTriangle(){
    return (totalAngle <= 180);
}
```

```
Function status: true
The shape is a VALID Triangle
```

Table 1-2. ILO B output observations and comments.

The output is determined based on the total angle of the given numbers, which are 40, 30, and 110. These numbers sum up to the number of 180, and so, it meets the condition that the total angle is at least 180 to be considered a valid triangle. This is because of the use of setting the total triangle and calling the functions to get the sum of the given angles.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO C

```
#include <iostream>
#include <cmath> // FOR USE IN POW
// FUNCTION DECLARATION
// ILO C #1
void swapTwoNum(int &a, int &b);
// ILO C #2
double kelvinToFahrenheit(double kelvin);
// ILO C #3
double calculateDist(double x1, double y1, double x2, double y2);
// ILO C #4
class Triangle{
    private:
        double sideA, sideB, sideC, angleA, angleB, angleC, totalAngle;

    public:
        Triangle(double sA, double sB, double sC, double A, double B, double C);
        void setValues(double sA, double sB, double sC, double A, double B, double C);

        const bool validateTriangle();
        double getArea();
        double getPerimeter();
        void triangleType();
};
```

```cpp
int main(){
    // ILO C #1
    int num1 = 15;
    int num2 = 30;

    std::cout << "Before Swap\n";
    std::cout << "num1 = " << num1 << std::endl;
    std::cout << "num2 = " << num2 << std::endl;
    swapTwoNum(num1, num2);
    std::cout << "\nAfter Swap\n";
    std::cout << "num1 = " << num1 << std::endl;
    std::cout << "num2 = " << num2 << std::endl;

    // ILO C #2
    double kelvin = 451;
    std::cout << "\nK Temperature in Fahrenheit: " << kelvinToFahrenheit(kelvin);
    // ILO C #3
    std::cout << "\nDistance Between Two Points: " << calculateDist(1, 3, 5, 9);
    // ILO C #4
    Triangle set1(10, 10, 10, 60, 60, 60);

    if(set1.validateTriangle()){
        std::cout << "\nThe shape is a valid Triangle." << std::endl;
        std::cout << "Area: " << set1.getArea() << std::endl;
        std::cout << "Perimeter: " << set1.getPerimeter() << std::endl;
        set1.triangleType();
    } else {
        std::cout << "\nThe shape is NOT a valid Triangle.\n";
    }

    return 0;
}
```

```cpp
// ILO C #1
void swapTwoNum(int &a, int &b){
    int swap = a;
    a = b;
    b = swap;
}

// ILO C #2
double kelvinToFahrenheit(double kelvin){
    return ((kelvin - 273.15) * 9/5) + 32;
}

// ILO C #3
double calculateDist(double x1, double y1, double x2, double y2){
    return std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));
}

// ILO C #4
Triangle::Triangle(double sA, double sB, double sC, double A, double B, double C){
    sideA = sA;
    sideB = sB;
    sideC = sC;
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}
```

```cpp
void Triangle::setValues(double sA, double sB, double sC, double A, double B, double C){
    sideA = sA;
    sideB = sB;
    sideC = sC;
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle(){
    return (totalAngle == 180);
}

double Triangle::getPerimeter(){
    return sideA + sideB + sideC;
}

double Triangle::getArea(){
    double s = getPerimeter() / 2;
    return std::sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}

void Triangle::triangleType(){
    if(angleA < 90 && angleB < 90 && angleC < 90){
        std::cout << "Triangle is ACUTE" << std::endl;
    } else if(angleA > 90 || angleB > 90 || angleC > 90){
        std::cout << "Triangle is OBTUSE" << std::endl;
    } else {
        std::cout << "Triangle is OTHER" << std::endl;
    }
}
```

Table 2-1. ILO C Code Program.

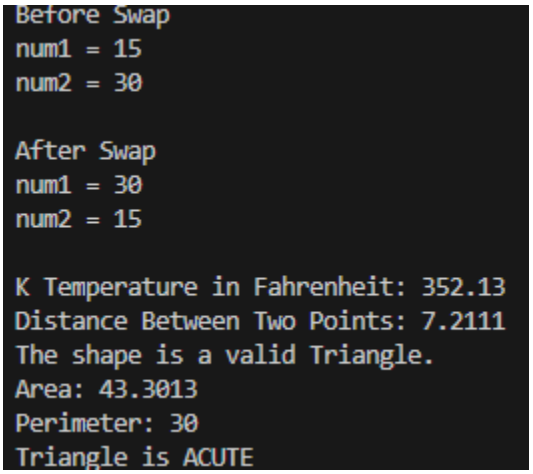


Table 2-2. ILO C output observations and comments.

The swapping of numbers for 15 and 30 uses reference operators in the function to take the given numbers, and assign them to the variables to process the swap operation. The numbers are printed to present the values before and after the swap operation. The conversion of temperature units from Kelvin to Fahrenheit involves using the function kelvinToFahrenheit to calculate the converted value, and returns the calculated value to the main function to display it. Cmath library is used for calculating exponents and square roots, which are demonstrated in getting the area of the sides..

CONCLUSION:

This Hands-on Activity has re-introduced many of the basic concepts of the C++ language, such as understanding the structure of the C++ code and its sections, the difference of local and global variables, and various operators and classes. Going to the procedures in ILO A and B, these parts demonstrated the use of functions, classes, and operators, such as boolean, logical, arithmetic, and relational operators. These operators are used for comparing the value of integers, determining the value of logical operations,and basic arithmetic operations, such as addition. Moving over to ILO C in the supplementary activities, these activities involve the use of private and public classes, using reference operators to easily swap integers, use of libraries, such as cmath to calculate square roots and exponents. Overall, this activity has brought back the fundamentals needed to write properly structured programs.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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